



HERZLICH WILLKOMMEN

International Energy Engineering | Solar Energy | Prof. Sirag Kartun

Group 1 :- 1. Ashish Manoj Mistari

2. Lakshmikantha Malode Ranganatharao

Agenda

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1. Introduction

Aim: “To design a cost-effective, efficient solar energy system”.

Objective :

- Analysis of the site
- Collection and Analysis of data
- Forecasting and designing of the solar system
- Economic calculations
- Selection of the solar system

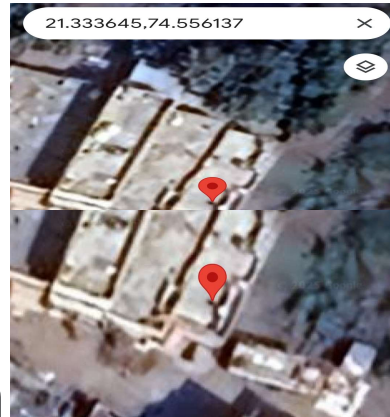
2. Geographical Location

Country :- India

City :- Pune, Maharashtra.

Location Coordinates:- 21°20'01.1"N 74°33'22.1"E

Building type :- 2 Floor Apartment



3. Building Specification

Type of Building :- Residential

Area :- 1200 m^2

Usable Area :- 900 m^2

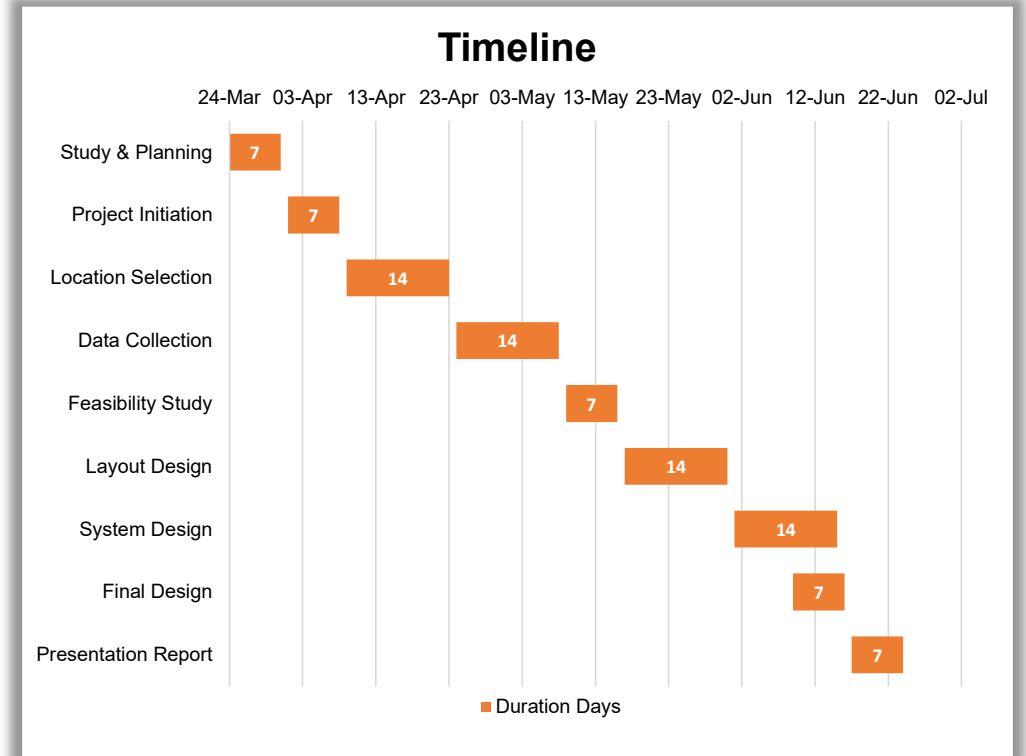
Orientation of Building :- South-West Facing



Source :- https://maps.app.goo.gl/uNLqEW6Sm1UzJp6r9?g_st=aw

4. Project Plan

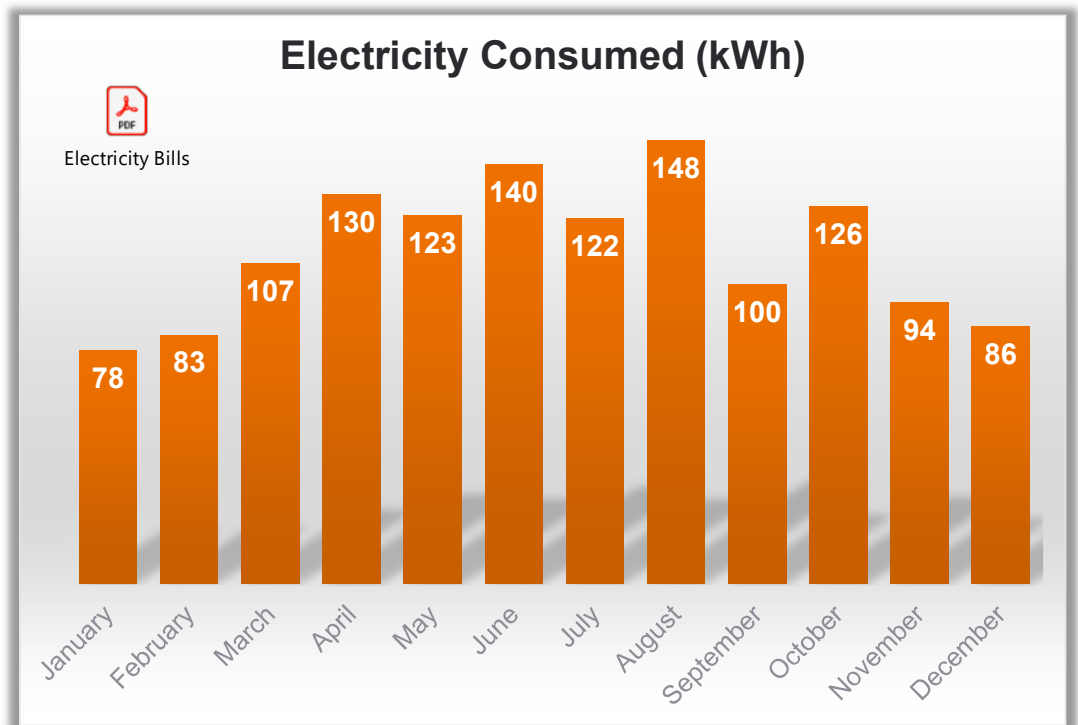
Task	Start Date	End Date	Duration Days	Progress
Study & Planning	24-Mar	31-Mar	7	Done
Project Initiation	01-Apr	08-Apr	7	Done
Location Selection	09-Apr	23-Apr	14	Done
Data Collection	24-Apr	08-May	14	Done
Feasibility Study	09-May	16-May	7	Done
Layout Design	17-May	31-May	14	Done
System Design	01-Jun	08-Jun	14	Done
Final Design	09-Jun	16-Jun	7	Done
Presentation Report	17-Jun	24-Jun	7	Done



5. Load Assessment

- Electricity Consumption data 2024

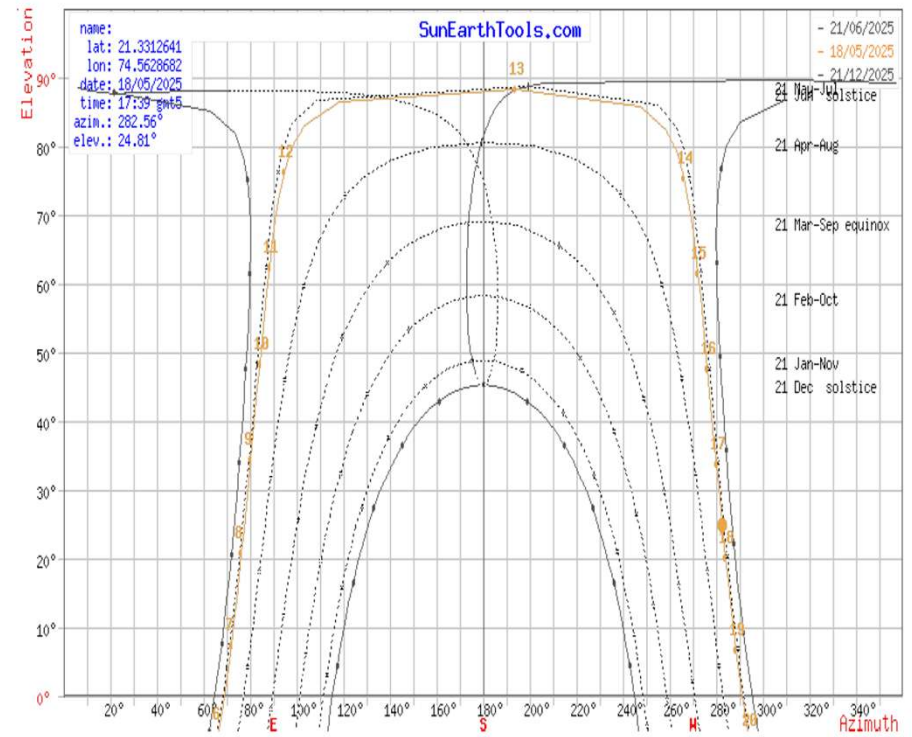
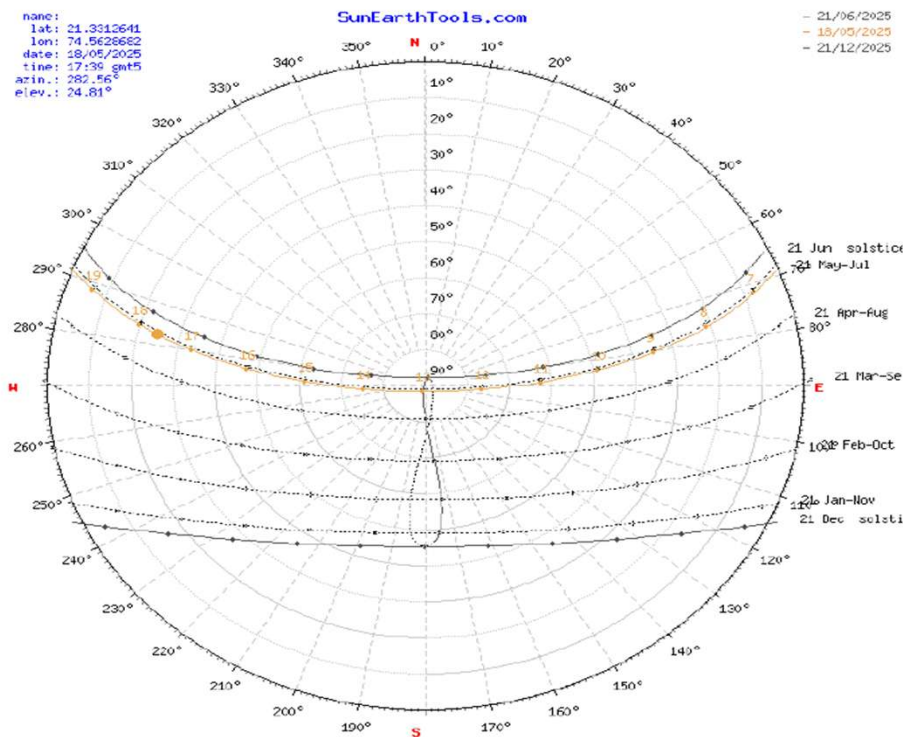
Month	Electricity Consumed (kWh)
January	78
February	83
March	107
April	130
May	123
June	140
July	122
August	148
September	100
October	126
November	94
December	86
Total	1337



Source:- Maharashtra State Electricity Distribution Company Limited Light Bills

6. Design of Layout

Sun Path Diagram

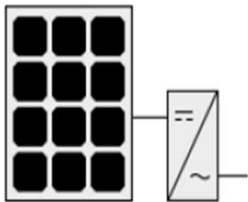


Source :- https://maps.app.goo.gl/uNLqEW6Sm1UzJp6r9?g_st=aw

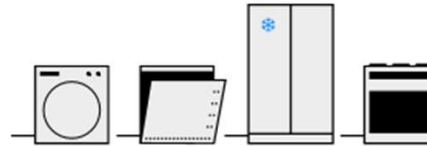
6. Design of Layout *(continued)*

Software :- Polysun

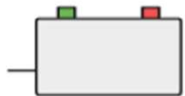
Project Project - System diagram 50f: Photovoltaics with battery



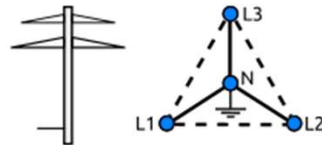
Photovoltaics: TP 300
Number of modules: 10
Total nominal power DC: 3 kW
Orientation (E=+90°, S=0°, W=-90°): 0 °
Tilt angle (hor.=0°, vert.=90°): 10 °



Number of electricity consumption profiles: 1
Consumption profile 1: Residential profile: 2 families
Total consumption of the profiles: 1,337 kWh



Battery: Li10 Set
Number of batteries: 1
Nominal capacity: 10.2 kWh



Electric Grid: Three-phase (127V/220V, 50 Hz, WYE)
Local grid voltage: 220 V

Feed-in power limitation: no

7. Comparison

Feature	Polycrystalline	Monocrystalline
Solar Panel Efficiency	Low Efficiency Rate 13% to 16%	High Efficiency Rate 16% to 23%
Appearance	Blue	Black
Cost	Less Expensive	More Expensive
Life Span	25-35 years	25-35 years
Temperature Coefficient	Low Temperature Coefficient	High Temperature Coefficient
Durability	Less Durable	Highly Durable
Roof Space	Requires more roof space	Works on Roofs with limited space

8. Parameter Specification

1. Local Electricity Grid

- Electricity Grid :- Three Phase (127V/220V, 50Hz,WYE)
- Grid Configuration :- Three Phase star (WYE)
- Nominal Voltage :- 220V
- Nominal Frequency :- 50Hz

2. PV Module

Photovoltaics	TP 300	
Manufacturer		Tata Power Solar Systems Ltd.
Data Source		Photovoltaikforum
Number of modules		10
Number of modules (layout)		10
Total nominal power DC	kW	3
Total gross area	m ²	19.84
Tilt angle (hor.=0°, vert.=90°)	°	10

Source :- Polysun Software

8. Parameter Specification *(continued)*

Tata Power Solar TP300

- Electrical Parameters at Standard Test Conditions (STC)

Nominal power output (W)	300
Power tolerance (W)	0 ~ +5
Module efficiency ($\eta\%$)	15.4
Voltage at P _{MAX} V _{MPP} (V)	36.6
Current at P _{MAX} I _{MPP} (A)	8.20
Open-circuit voltage V _{oc} (V)	44.8
Short-circuit current I _{sc} (A)	8.71

- Electrical Parameters at NOCT

Power Output P _{max} (W)	216.0
Voltage at P _{MAX} V _{MPP} (V)	32.2
Current at P _{MAX} I _{MPP} (A)	6.71
Open-circuit voltage V _{oc} (V)	39.4
Short-circuit current I _{sc} (A)	7.24

8. Parameter Specification *(continued)*

Tata Power Solar TP300

- Temperature coefficient characteristics

NOCT (°C)	47 ± 2
Module efficiency (%/°C)	-0.06 ± 0.01
Temperature coefficient of P _{MAX} (%/°C)	-0.41
Temperature coefficient of V _{OC} (%/°C)	-0.32
Temperature coefficient of I _{SC} (%/°C)	0.058

- Operating conditions

Maximum system voltage (UL & IEC) (V)	1000 & 1000
Maximum series fuse rating (A)	20
Limiting reverse current (A)	20
Operating temperature range (°C)	-40 and +85
Maximum static load (snow or wind)	113 psf (5400 Pa)

8. Parameter Specification *(continued)*

3. Inverter

Inverter 1: Name		MIC 3300TL-X
Inverter 1: Manufacturer		Growatt New Energy Co., Ltd
Layout 1: Number of inverters		1
Total nominal power AC	kVA	3.3
Energy production DC [Qpvf]	kWh	4,460
Energy production AC [Qinv]	kWh	4,246

- Input Data :-

Max. recommended PV power (for module STC)	4290W
Max. DC voltage	550V
Start voltage	80V
MPP voltage range	65V-550V
Nominal voltage	360V
No. of PV strings per MPP tracker	1



8. Parameter Specification *(continued)*

Growatt MIC 3300TL-X2

- Output data (AC)

AC nominal power	3300W
Max. AC apparent power	3300VA
Nominal AC voltage(range*)	230V (180-280V)
AC grid frequency(range*)	50/60 Hz (45-55Hz/55-65 Hz)
Max. output current	14.3A
Adjustable power facto	0.8leading...0.8lagging
THDi	<3%
AC grid connection type	Single phase

- Efficiency

Max. efficiency	97.6%
European efficiency	97.1%
MPPT efficiency	99.9%

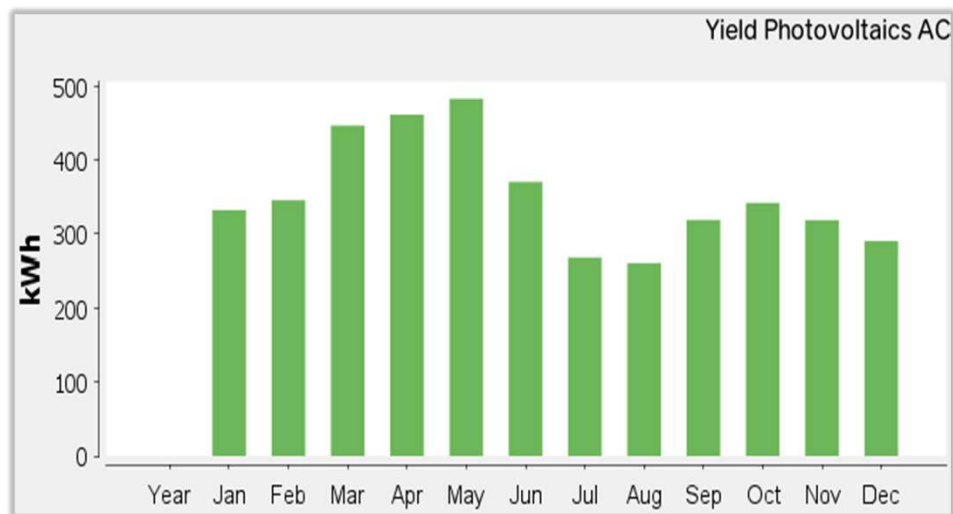
8. Parameter Specification *(continued)*

4. Battery

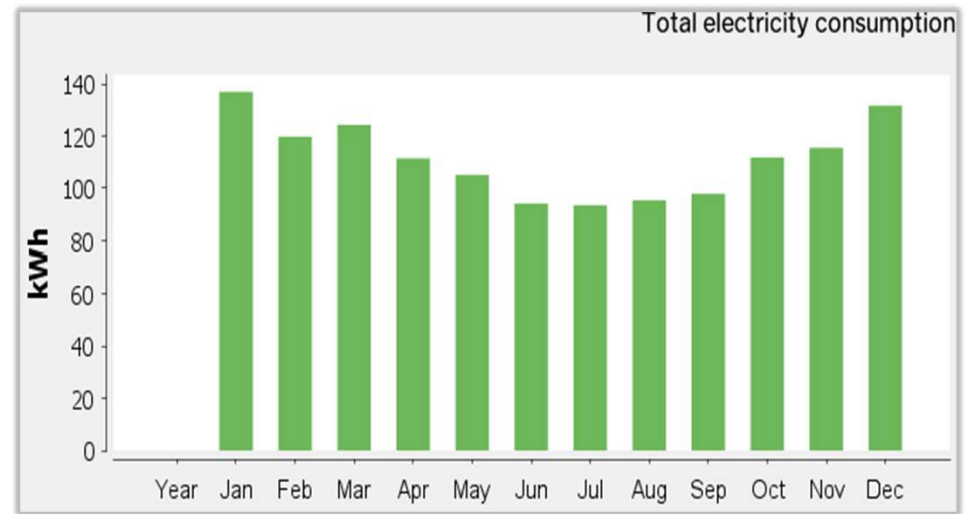
Battery	Li10 Set	
Number of batteries		1
Nominal capacity	kWh	10.2
Cumulative damage [D]	%	0.8
Lifetime in years	a	20

9. Result

- Electricity Produced by **PV Modules**



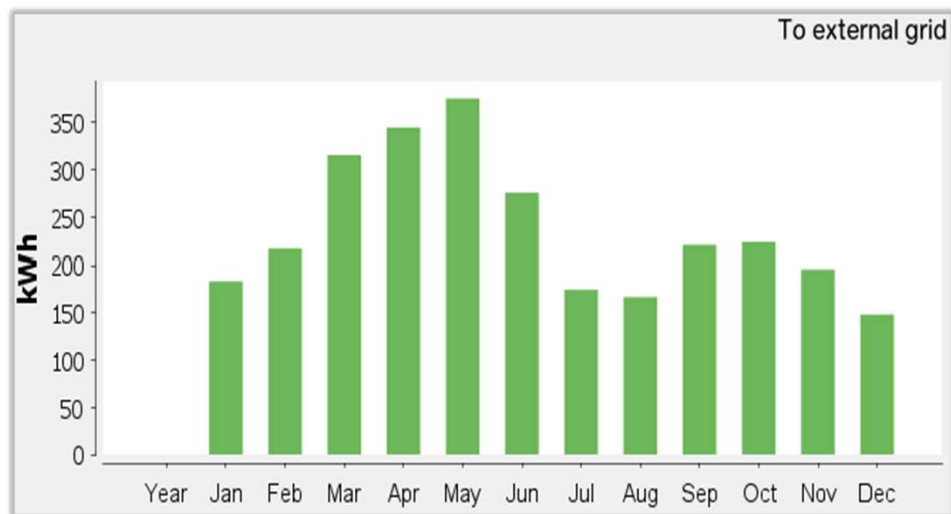
- Electricity Consumption



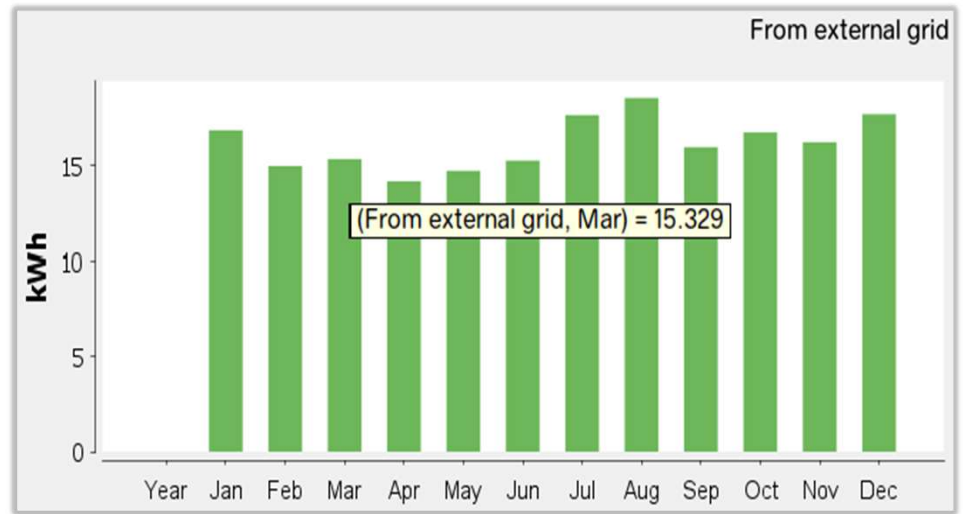
Source :- Polysun Software

9. Result *(continued)*

- Electricity supplied to the Grid



- Electricity from the Grid



10. Feasibility Study

Funding Schemes under Government of India

- **Pradhan Mantri Surya Ghar Yojana (PMSG)**
 - Support benefits 30000 ₹ per kilowatt upto 2 kW power system.
 - Additional support 18000 ₹, if system power goes above 3 kW.
 - Maximum 78000 ₹ will be supported from Government of India.
 - Low interest loan financing from the Bank at 7 % interest rate under 3 kW power PV system.
 - These incentive schemes are only for residential households.

10. Feasibility Study *(continued)*

Investment Parameters

PV Module cost	8700.00 ₹/Module
Number of Module	10.00 Module
Total PV Module cost	87000.00 ₹
Inverter Cost	19500.00 ₹
Battery Cost	252000.00 ₹
Installation Cost	30.00 ₹ / W
Total installation Cost	90000.00 ₹
Total Lifespan	27.00 years

Polysun Data_with battery

Yield Photovoltaics	4245.00 kWh
Electricity Demand	1337.00 kWh

Polysun Data_without battery

Yield Photovoltaics	4469.00 kWh
Self Consumption	693.00 kWh

Energy cost

From grid	10.00₹
To grid	3.00₹

10. Feasibility Study *(continued)*

1. CAPEX (without incentive)

CAPEX Calculation with Battery			
Appliances	No. of Units required	Per unit cost (₹)	Total Cost (₹)
PV Modules	10.00	8700.00	87000.00
Battery	1.00	252000.00	252000.00
Inverter	1.00	19500.00	19500.00
Installation per W		9000.00	90000.00
Total CAPEX			448500.00

CAPEX Calculation without Battery			
Appliances	No. of Units required	Per unit cost (₹)	Total Cost (₹)
PV Modules	10.00	8700.00	87000.00
Inverter	1.00	19500.00	19500.00
Installation		9000.00	90000.00
Total CAPEX			196500.00

2. CAPEX (with incentive)

CAPEX Calculation_with_battery	370500.00 ₹
CAPEX Calculation_without_battery	118500.00 ₹

Source:- PV module Cost Access Time & Date 11.30 19062025 , Growatt MIC3300TL-X Access Time & Date 11.45 19062025

10. Feasibility Study *(continued)*

3. OPEX

OPEX Calculation with Battery	
Maintenance Cost	13455.00 ₹ / a

OPEX Calculation without Battery	
Maintenance Cost	5895.00 ₹ / a

4. LCOE

LCOE_with_Battery	6.40 ₹ / kWh
LCOE_without_Battery	2.30 ₹ / kWh

10. Feasibility Study *(continued)*

Condition :- 1 (with Battery 100% Feed In)			
Parameter	kWh	₹ / a	€ / a
Self Consumption	-	-	-
Elec. From External Grid	1337.00	13370.00	134.82
Feed-in tariff	4245.00	12735.00	128.42
Savings	-	-635.00	-6.40

Condition :- 2 (With Battery 33% consumption based on demand)			
Parameter	kWh	₹ / a	€ / a
Self Consumption	441.21	4412.10	44.49
Elec. From External Grid	895.79	8957.90	90.33
Feed-in tariff	3803.79	11411.37	115.07
Savings		6865.57	68.65
Payback Period	53 years		

10. Feasibility Study *(continued)*

Condition :- 3 (With Battery 66% consumption based on demand)			
Parameter	kWh	₹ / a	€ / a
Self Consumption	882.42	8824.20	88.98
Elec. From External Grid	454.58	4545.80	45.84
Feed-in tariff	3362.58	10087.74	101.72
Savings		14366.14	143.66
Payback Period	25.7 years		

Condition :- 4 (Without Battery)			
Parameter	kWh	₹ / a	€ / a
Self Consumption	693.00	6930.00	69.88
Elec. From External Grid	644.00	6440.00	64.94
Feed-in tariff	3776.00	11328.00	114.23
Savings		11818.00	118.18
Payback Period	10 years		

Thank You

Professional Report Without Battery

	Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Yield Photovoltaics DC [Qpvf]													
kWh	4691	402	404	495	493	498	379	280	277	345	387	377	352
Radiation onto module area [Esol PV]													
kWh	38374	3214	3276	4091	4108	4160	3119	2291	2263	2807	3174	3053	2818
Yield Photovoltaics AC [Qinv]													
kWh	4469	384	386	473	471	476	361	265	262	328	368	360	336
Total electricity consumption [Ecs]													
kWh	1337	137	120	124	111	105	94	93	95	98	112	115	132
Self-consumption [Eocs]													
kWh	693	66	60	67	61	60	55	52	50	51	58	55	59

Professional Report With Battery

	Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Yield Photovoltaics DC [Qpvf]													
kWh	4460	349	363	469	484	507	390	284	278	336	360	335	306
Radiation onto module area [Esol PV]													
kWh	36408	2777	2925	3853	4026	4228	3204	2319	2267	2729	2942	2699	2439
Yield Photovoltaics AC [Qinv]													
kWh	4246	333	346	448	463	484	371	268	263	319	342	319	290
Total electricity consumption [Ecs]													
kWh	1337	137	120	124	111	105	94	93	95	98	112	115	132
Self-consumption [Eocs]													
kWh	1405	149	129	132	118	109	95	93	96	99	119	123	142